Reeth S Kawad

213-512-8528 | reethkawad@gmail.com | LinkedIn | Portfolio

EDUCATION

University of Southern California, Viterbi School of Engineering Los Angeles, CA
B.S. Mechanical Engineering *May 2026*
Relevant Coursework: Linear Control Systems, Turbine Design and Analysis, Heat Transfer, Mechanics of Materials, Fluid Dynamics

WORK EXPERIENCE

GrayMatter Robotics, *Robotics Systems and Applications Intern* *Jan 2026-May 2026*

- Built a **multimodal DAQ pipeline** integrating force, IMU, position trackers, and thermal sensors for **hardware-in-the-loop (HIL)** characterization of human sanding processes, reducing industrial engineering time studies by 40%
- Spearheaded system and application engineering for customer parts (e.g. helmets), conducting reachability studies, 3D scanning, **point-cloud segmentation (**MeshLab**)**, and **toolpath planning (ROS2),** ensuring GitHub version control

Lumindt Labs, *Mechanical Engineering Intern* *Jun 2025-Aug 2025*

- Designed and fabricated a thermal test fixture in **SolidWorks** housing a transient hot-wire sensor; wrote **Python DAQ (Raspberry Pi)** to measure thermal conductivity of four metal powders (<10% uncertainty), enabling material down-selection
- Built an **Excel-based** heat exchanger and **coolant loop calculator** to evaluate efficiency for system-level thermal management
- Supported process engineering by selecting BOP components and conducting **ASME standard**-based preliminary stress analyses

USC Baum Family Makerspace, *Machinist and Fabrication Engineer* *Sep 2023-Dec 2025*

- Machined **GD&T** and precision (±1 thou) components on Haas CNC Mill/Lathe, ProtoTRAK, and Omax waterjet; interpreted **technical drawings** and optimized toolpaths in **MasterCam**
- Fabricated **3D printed** prototypes (PLA, resin, carbon fiber) for 12+ design teams and 80+ research labs, applying design-for manufacturing (**DFM**), including aerodynamic elements for Formula SAE

USC Dynamic Robotics and Controls Laboratory, *Mechanical Engineer* *Sep 2025-Dec 2025*

- Designed and rapid-prototyped a **series elastic actuator (SEA**) for an N20 motor (**SolidWorks**), benchmarking force feedback fidelity for a **robotic finger** against design targets

FireWarden, *Cofounder and CTO* *Jan 2025-Sep 2025*

- Co-founded a **wildfire protection startup;** developed a calculator analysing home dimensions to recommend wildfire defence systems rated for 8+ hours of autonomous operation, with first product being a **pool-sourced sprinkler system**
- Secured **$20,000 prize funding** by winning DAS InnovateLA Pitch Competition and enrolled into TechStars Pre-accelerator

LEADERSHIP

Collegiate Wind Competition, *Team Lead* *Sep 2025-May 2026*

- Designed end-to-end **electrical schematics** for turbine controls, integrating RPM, voltage, and current sensors via **I2C communication** on an **Arduino UNO;** wrote firmware**,** performed soldering, and debugged sensor communication
- Engineered an **electric braking system** with power resistors as a variable resistive load for rated-power control **and safe turbine shutdown**, verified through **bench instrument testing** (multimeter, oscilloscope)
- Guided mechanical design of turbine; ran blade aerodynamic simulations (**QBlade**) to optimize geometry for target power
- Led **project development** sub-team to devise a utility-scale wind farm, performing site assessment, grid interconnection analysis, and financial and policy planning to meet DOE competition deliverables

Formula SAE (SC Racing), *Aerodynamics and Vehicle Dynamics Engineer* *Jan 2023-Apr 2025*

- Streamlined aerodynamic meshing using a fault-tolerant mesh system in **ANSYS Fluent**, and automation by **MATLAB scripting**
- Developed **setup and validation tools**, like **toe alignment tool** and **yaw probe**, to verify simulations against real-world testing

PROJECTS

Adaptive Pitch Control for a Vertical Axis Wind Turbine

- Implemented a **collective blade pitch control system** for a 5” H-type VAWT, increasing aerodynamic efficiency by ~10% via an optimal angle lookup table derived from **wind tunnel-tested** CoP data

Performance Analysis of Honeycomb Flow Straightener in a Wind Tunnel

- Modelled, fabricated, and tested a 1.5m blow-down wind tunnel, validating a honeycomb flow straightener design that reduced turbulence by 53% at front and 86% at back of test space, using **Siemens NX for CAD and Excel for data analysis**